When no more ammonia passes over (test the distillate with red litmus) the receiver is removed. The excess of acid is then determined by titration with alkali, using phenolphthalein as the indicator.

Calculation:

Weight of the substance = Wg.

Volume of H_2SO_4 required for the complete neutralisation of evolved NH_3 = V mL.

Strength of H_2SO_4 used to neutralise NH_3 = N

Let the Volume and the strength of NH_3 formed are V_1 and N_1 respectively

we know that $V_1N_1 = VN$

The amount of nitrogen present in the w g of Organic Compound = $\frac{14 \times NV}{1 \times 1000 \times w}$

Percentage of Nitrogen = $\left(\frac{14 \times NV}{1000 \times w}\right) \times 100\%$

Example : 0.6 g of an organic compound was Kjeldalised and NH_3 evolved was absorbed into 50 mL of semi-normal solution of H_2SO_4. The residual acid solution was diluted with distilled water and the volume made up to 150 mL. 20 mL of this diluted solution required 35 mL of $\frac{N}{20}$ NaOH solution for complete neutralization. Calculate the % of N in the compound.

Weight of Organic compound = 0.6g
Volume of sulphuric acid taken= 50mL
Strength of sulphuric acid taken= 0.5 N
20 ml of diluted solution of unreacted sulphuric acid was neutralised by 35 mL of 0.05 N Sodium hydroxide

Strength of the diluted sulphuric acid = $\frac{35 \times 0.05}{20}$

= 0.0875 N

Volume of the sulphuric acid remaining after reaction with Organic compound = V_1 mL

Strength of H_2SO_4 = 0.5N

Volume of the diluted H_2SO_4 = 150 mL

Strength of the diluted sulphuric acid = 0.0875 N

$V_1 = \frac{150 \times 0.087}{0.5}$ = 26.25 mL

Volume of H_2SO_4 consumed by ammonia = 50 - 26.25

= 23.75 mL

23.75 mL of 0.5 N $H_2SO_4 \equiv$ 23.75mL of 0.5N NH_3

The amount of Nitrogen present in the 0.6 = $\frac{14g}{1000\ mL \times 1\ N} \times 23.75 \times 0.5N$

= 0.166g

Percentage of Nitrogen = $\frac{0.166}{0.6} \times 100$

= 27.66 %

Evaluate Yourself

10) 0.3 g of an organic compound on kjeldahl's analysis gave enough ammonia to just neutralize 30 mL of 0.1N H_2SO_4. Calculate the percentage of nitrogen in the compound.

Unit 11 revised - jagan.indd 148 22-01-2021 12:42:31